ADRIÀ MITJAVILA VENTURA

MSc | PhD student in Bioinformatics at *Institut Josep Carreras*

Biotechnologist with training in Molecular Genetics and Bioinformatics. Interested in omics data analysis to study the mechanisms of gene regulation and their evolution. See more about me on my website.



EXPERIENCE

11/2022 | Present

PhD student

Institut Josep Carreras & *Institut Germans Trias i Pujol* — Badalona, Spain

Group of Regulatory Genomics (Dr. Tanya Vavouri)
Study of the role of transposable elements in the evolution of gene regulation in mammals.

- Analysis and integration of small RNA-seq, RNA-seq and genomics data.
- Mentoring of graduate students and collegues.
- Workshops addressed to high-school students.

08/2021 | 11/2022

Bioinformatician & master's thesis student

Institut Josep Carreras — Badalona, Spain

Group of Regulatory Genomics (Dr. Tanya Vavouri)
Study of Piwi-interacting RNAs (piRNAs), their biogenesis mechanisms and their conservation across species.

- Analysis and integration of small RNA-seq, RNA-seq and genomics data.
- Master's thesis (UOC): *Analysis of the variation in piRNA expression across three Mus species*.
- Mentoring of bachelor's and master's thesis students.
- Workshops addressed to high-school students.

07/2019 | 07/2021

Bioinformatician

Istituto Europeo di Oncologia — Milan, Italy

Group of Epigenetic Mechanisms in Cancer (Dr. Diego Pasini)
Study of Wnt/βcatenin signaling pathway -and others- and its effects in transcription and epigenetic modifications in the context of colorectal cancer and stem cell development.

- Analysis and integration of omics data (RNA-/ChIP-/ATAC-seq).
- Comparison of different ATAC-seq peak callers.
- Comparison of different sequencing depths for RNA-seq and their influence in differential expression analysis.

09/2017 | 10/2018

Master's thesis student & Research assistant

Instituto de Biomedicina de Sevilla — Seville, Spain

Group of Gene Expression (Dr. Sebastián Chávez)
Study of Xrn1 and Ccr4 roles in yeast nucleosome architecture and gene transcription.

- Analysis of MNase-seq data and MNase-qPCR experiments from Xrn1-, Ccr4-mutant and WT yeast strains.
- Master's thesis (US): *Influence of Xrn1 in nucleosome positioning […] and its effects in transcription*.

07/2017 | 09/2017

Internship student

Institut d'Investigació Biomèdica de Girona — Girona, Spain

Unit of Diabetes, Endocrinology and Nutrition
Extracurricular internship. Identification of novel factors (genetic and bacterial) related to adipose tissue physiology and obesity.

- RNA/DNA extraction from mouse hepatic biopsies and human fecal material.

CONTACT

Girona (Spain)
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SKILLS

Bioinformatics:

sRNA/RNA/ChIP/ATAC/MNase-seq data analysis, Sequence analysis, Genome Browsers (UCSC, ENSEMBL, IGV)

Informatics:

R/Bioconductor, RMarkdown, Shiny, Bash, Snakemake, Python, Git, HPC (PBS/Slurm), Conda, Docker, Singularity, Linux (Ubuntu)

Languages:

Catalan (Native), Spanish (Bilingual), English (IELTS 7.5 - C1), Italian (Basic)

DEVELOPMENT

`plotmics`: visualization of omics and sequencing data in R.

`ggmitji`: complements and customization for ggplots.

`chromHMMviewR`: visualization of ChromHMM outputs in R.

More on my GitHub profile.

References available upon request.
*Build with **pagedown**, code here.*
Last updated on 2025-11-13.

09/2016 | 06/2017

Internship & bachelor's thesis student

Microbial. Sistemes i aplicacions analítiques, S.L. — Girona, Spain

Microbiological and molecular analyses of food and water for pathogen detection and food forensics | Study of butyrate effects on the ileal microbiota of colectomized colorectal cancer patients.

- DNA extraction of intestinal mucosa biopsies.
- (q)PCR experiments to detect clusters and families of bacteria.
- Statistical analysis of NGS data to quantify bacterial genera in intestinal biopsies.
- Bachelor's thesis (UdG): *Butyrate-induced changes in the […] microbiota in colorectal cancer patients […].*

EDUCATION

2022 | Present

PhD in Bioinformatics

Universitat Autònoma de Barcelona (UAB) — Barcelona, Spain

- INPhINIT Incoming PhD Fellowship from LaCaixa.
- Research carried out at Regulatory Genomics group of the Institut Josep Carreras (IJC).
- Supervisor: Dr. Tanya Vavouri.

2018 | 2022

MSc in Bioinformatics and Biostatistics

Universitat Oberta de Catalunya (UOC) — Online

- Focused on Computational Genomics.
- Master's thesis: *Analysis of the variation in piRNA expression across three close related Mus species*. Grade: 9.7/10 (*with honours*). Supervisor: Dr. Tanya Vavouri.
- Overall grade: 8.94/10.

2017 | 2018

MSc in Molecular Genetics and Biotechnology

Universidad de Sevilla (US) — Seville, Spain

- Specialization in Molecular Biology Research.
- Master's thesis: *Influence of Xrn1 in nucleosome positioning across Saccharomyces cerevisiae genome and its effects in transcription*. Grade: 9.2/10. Supervisors: Dr. Sebastián Chávez de Diego & Dr. Gabriel Gutiérrez Pozo.
- Overall grade: 9.33/10.

2012 | 2017

BSc in Biotechnology

Universitat de Girona (UdG) — Girona, Spain

- Specialization in Cellular & Molecular Biotechnology.
- Bachelor's thesis: *Butyrate-induced changes in the diversity of intestinal mucosa-associated microbiota in colorectal cancer patients submitted to a lateral ileostomy*. Grade: 9.4/10. Supervisor: Prof. Jesús García-Gil.
- Overall grade: 8.12/10.

PUBLICATIONS

2025

Genetic polymorphisms lead to major, locus-specific, variation in piRNA production in mouse
Casas, Mitjavila-Ventura *et al.*, *EMBO Journal*, 2025. doi: 10.1038/s44318-025-00475-4

2021

Xrn1 influence on gene transcription results from the combination of general effects on elongating RNA pol II and gene-specific chromatin configuration
Begley *et al.*, *RNA Biology*, 2021. doi: 10.1080/15476286.2020.1845504

POSTERS & PRESENTATIONS

12/2024

Genetic polymorphisms lead to major, locus-specific, variation in piRNA production in mouse
Mitjavila-Ventura, Casas et al. Poster at the *Jornades de Bioinformàtica i Genòmica* from the *Societat Catalana de Biologia*.
Lleida, Spain

05/2023

Fast evolution of piRNA cluster expression in mouse.
Mitjavila & Vavouri. Poster at the Microsymposium in RNA Biology. — Vienna, Austria

HONORS & AWARDS

2022 | 2025 — **INPhINIT Incoming PhD Fellowship**
La Caixa Foundation

2022 — **Master's thesis**, ***with honours***
Universitat Oberta de Catalunya

2019 | 2021 — **SEMM PhD Fellowship**
Scuola Europea di Medicina Molecolare (declined)

2015 | 2021 — **Elite athlete**
Boletin Oficial del Estado

2016 — **World Rowing U23 Champion**
FISA - World Rowing

ORGANIZATIONS

2018 | Present — ***Societat Catalana de Biologia*** (SCB)
Member (2018-Present)

2017 | Present — ***Associació de Biotecnòlegs de Catalunya*** (ASBTEC)
Member (2017-Present), collaborator (2018-Present) and board member (2019-Present)